



To research about open source model:

In AI

1. How many open source models are there?
2. How many closed source models are there?
3. Which models are good?
4. Suppose if you have 16 GB of RAM and 256GB hard disk, which model can you run locally, open source?

There isn't a single exact count for "open-source" or "closed-source" AI models because the numbers change constantly and the categories aren't standardized. A current public model hub shows 2,904,662 models listed on Hugging Face, which is a good indicator of scale for publicly shared models, and another source says closed models account for about 80% of model usage.^[1] ^[2]

How many models exist

Open models are counted by public repositories, while closed models are usually not publicly listed in the same way, so a precise total for either group is not available from one reliable global source. Hugging Face itself shows 2,904,662 models on its platform, but that includes many kinds of models, not only chat LLMs.^[2] ^[1]

Which models are good

For general use, the strongest open-weight families in 2026 are commonly listed as Llama, Qwen, Mistral, and DeepSeek variants, with open models now reaching near-frontier quality on many tasks. For local use, popular good options include Llama 3.1 8B, Qwen 2.5 7B, and DeepSeek-R1 distilled 8B, especially if you want a balance of quality and speed.^[3] ^[4] ^[5] ^[2]

What you can run locally

With 16 GB RAM and 256 GB disk, you can comfortably run quantized 7B–8B models, and often some 14B models in smaller quantizations if you are careful with settings. A source aimed at 16 GB systems lists Llama 3.1 8B, Qwen 2.5 7B, Mistral 7B, DeepSeek R1 8B, and Qwen 2.5 14B as viable local choices, with the 8B-class models being the safest bet.^[6] ^[3]

Practical picks

If you want the simplest recommendation:

- Best overall on 16 GB: Llama 3.1 8B or Qwen 2.5 7B. ^[4] ^[3]
- Best for reasoning: DeepSeek-R1 distilled 8B. ^[3] ^[4]
- Best for multilingual work: Qwen 2.5 7B. ^[4]
- If you want to push higher: Qwen 2.5 14B in a quantized version, but expect slower performance. ^[6] ^[3]

Simple rule

For your machine, think of 8B as the sweet spot and 14B as the upper limit for local open models, depending on quantization and how much RAM the rest of your system uses. If you want reliable day-to-day chatting, coding help, or writing assistance, start with an 8B model before trying larger ones. ^[5] ^[3] ^[6] ^[4]

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